

The Reputation *Gap.*

A scrollytelling story about EA FC 25 player ratings and the 2024-25 season that followed them. We line up the in-game overall against what players actually produced on the pitch, and call the distance the Reputation Gap.

TEAM

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LIVE SITE <https://com-480-data-visualization.github.io/Tito-Data-Viz/>

Every summer, a single number sets the tone for an entire football year.

EA's Overall Rating is one of the most widely cited football statistics outside the analytics community, and one of the few no outside observer can actually audit. It blends last season's output with positional weights, league modifiers and an implicit reputation component. We wanted to know how much of the rating is the player, and how much is the story we have decided to tell about them.

Once the 2024-25 season was done, which players actually *justified* their EA FC-25 rating, and which ones either coasted on reputation or deserved *far better*?

RESEARCH QUESTION

The question is easy to ask and difficult to answer cleanly. A striker who scores twenty league goals is not produced by the same system as a centre-back who blocks eighty shots. Comparing them needs a frame: a way to read every player against the people who play their actual job. The whole project is built around finding that frame, then showing it three times in a row, each time at a closer zoom.

WHAT WE END UP MEASURING

A directional residual, per sub-position.

For every sub-position we fit a linear regression of our real-performance composite on EA's OVR. The Reputation Gap is the residual. Positive means the player out-produced their reputation, negative means the reputation is out-producing them.

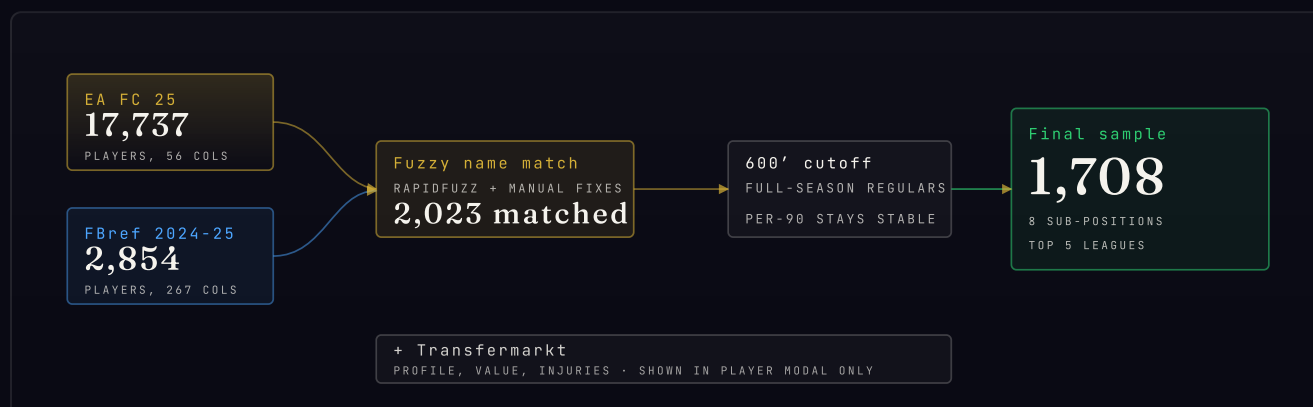
AUDIENCE

Fans first, analysts second.

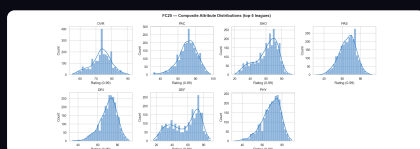
The story has to land for someone who plays FC 25 on a Friday night and has never seen the letters xG before, and still be sharp enough for the analyst who lives in those numbers all week. Most of the design work hides behind that double brief.

Two main sources, one residual.

17,737 → 2,023 → 1,708.

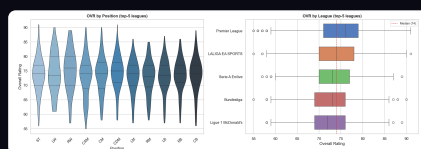


M1 • EDA, EXPANDED



COMPOSITE DISTRIBUTIONS

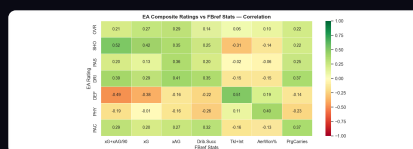
DEF variance is enormous ($\sigma 17.9$), a single global metric would crush defenders. → composite must normalise per position.



OVR BY POSITION & LEAGUE

IQR bands shift from league to league. → one regression line per sub-position, refit on every filter switch.

THREE CHARTS THAT EACH CLOSED A DOOR



EA ↔ FBREF CORRELATIONS

SHO ↔ xG and DEF ↔ TKI + Int strongest; PAC ↔ ProgCarries weak. → five dimensions, not seven.

92.9%

MATCH RATE ON
REGULARS ($\geq 600'$)

600'

LEAGUE-MINUTES
CUTOFF

5

PERFORMANCE
DIMENSIONS

8

SUB-POSITIONS, GK TO
ST

● METHOD, IN PLAIN ENGLISH

- Inside each sub-position, every relevant FBref stat is percentile-ranked across the players in that role. A central midfielder is judged against other central midfielders, not against strikers.
- Percentiles are grouped into five dimensions, scoring, creation, progression, defense and discipline, and combined with sub-position weights. The result is rescaled so the mean and spread match EA's OVR (about 75 ± 5), keeping the two numbers directly comparable.
- For every sub-position we then fit a linear regression of composite on OVR. The Reputation Gap is the signed residual: above the line is underrated, below is overrated.

PERCENTILE INSIDE, Z-SCORE OUTSIDE

A striker's npG is not a centre-back's npG.

Raw values for the same stat sit on different distributions depending on the role. Percentile-ranking inside the sub-position keeps the channel consistent and stops one role's heavy tail from drowning out another. The weighted composite is then z-scored to align with EA's OVR scale.

FILTER EVOLUTION • 300' → 600'

M1 kept 2,168 names at 300'. M2 hardened to 600'.

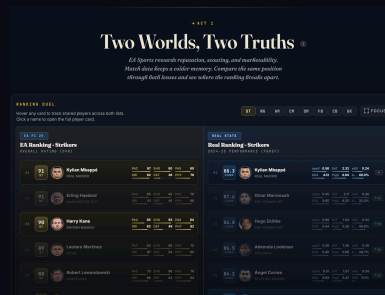
At 300' per-90 metrics stayed spiky for the long tail, one cup cameo distorted npG/90 for back-up forwards. The 600' cut takes us from 2,168 to 1,708 names. We lose 460 rows but every dot on the scatter has genuine season-long exposure.

From a flat scatter to a rebuild. *Each act was born differently.*

● ACT I • TWO WORLDS

The first instinct was a single ranked delta list: name, EA, real, gap. We dropped it within an afternoon. The opening should *not* tell the reader who is wrong, it should put both rankings on the same shelf and let the disagreement do the talking. *Gold* versus *blue* carries the dispute, the sub-position toggle keeps it fair, animated lines bridge crossover players. A centred "gap %" label was tried and removed: it made the disagreement feel non-negotiable.

VIZ CHOICE → LINKING LINES FOR CROSSOVER NAMES



SPLIT RANKING STRIKERS,
SAME SUB-POSITION



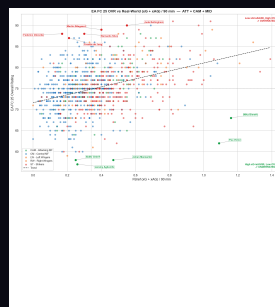
PLAYER CARD SUB-RATINGS •
COMPOSITE • DIMENSIONS

● ACT II • THE GAP, PLOTTED

M1 sketched OVR against $xG + xAG/90$ with one global line. It worked for attackers and broke for everyone else: a defender's $0.05 xG/90$ is not a striker's 0.05 . The y-axis became a five-dimension composite, rescaled to share OVR's spread, with a regression line refit *per sub-position*.

Two priming cards open the act, Griezmann and Marmoush. The reader votes before the chart loads, so by the time the scatter lands it reads as agreement or disagreement, not a cold reveal.

VIZ CHOICE → HUE FOR POLARITY, AREA FOR MINUTES



M1 SKETCH ONE
GLOBAL LINE



M2 FIRST PASS PER-
POSITION LINE,
SIDEBAR



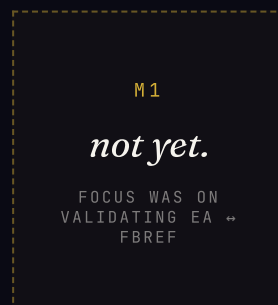
M3 LIVE LINKED
WALLS, MAP, STADIUM

● ACT III • REBUILD THE RATINGS

Not planned in M1. The idea emerged once we accepted our composite hides a value choice. M2's first pass was pick, quiz, tune, test, but dropped the reader on raw sliders, punishing for anyone without an opinion to start from.

M3 puts the quiz upstream so the sliders open *pre-tilted*. The live top-ten reshuffles in place with arrows from the EA baseline, every tug has an immediate consequence, and the taste test at the end checks whether the gut matches the weights.

VIZ CHOICE → TERNARY PLOT FOR TASTE, LIVE TOP-TEN WITH MOVERS



M1 CONCEPT NOT
PLANNED



M2 FIRST
PASS COLD-START
SLIDERS



M3 LIVE QUIZ PRE-
WARMS THE SLIDERS

How we chose to tell it.

Every judge has a bias.

● THE STORY WE WANTED TO TELL

NO ONE IS REALLY RIGHT • RATINGS DEPEND ON THE JUDGE

THE THESIS

EA's bias *is real.*

Across 1,708 players the signed residual to our composite is large and clearly directional inside several sub-positions. The numbers tell a story EA does not. Act II is built to land that fact first.

THE CATCH

Our composite *is opinionated.*

We picked the dimensions. We picked the weights. A different analyst would have drawn a different line and called different players overrated. The story refuses to claim one final truth.

THE HAND-OVER

The reader *is the judge.*

Act III turns the weight choice into the interaction. By the last screen the reader has built their own composite, seen their own residuals, and learned that the gap is as much about them as about EA.

● ART DIRECTION

TYPE

Fraunces *ø' Inter.*

Fraunces display, Inter body, JetBrains Mono stats. Italic axes carry every act header.

Aa Aa 75.5

COLOUR

Gold on *ink.*

Accent #d4af37, green / red for gap polarity, blue for context.



MOTION

Scroll, *don't shout.*

Scrollama drives the acts, GSAP the ball menu, D3 every chart rebuild.

SCROLLAMA

GSAP

D3

● ENCODING CHOICES ON THE SCATTER

THREE CHANNELS, ONE RESIDUAL

POSITION • SIGNED RESIDUAL

Each dot sits at (OVR, real composite). The y-axis distance to the per-position line is the gap. Above the line, underrated; below, overrated.

HUE • GAP POLARITY

Green underrated, red overrated, grey near zero. Polarity sits on the residual, never on the raw value.

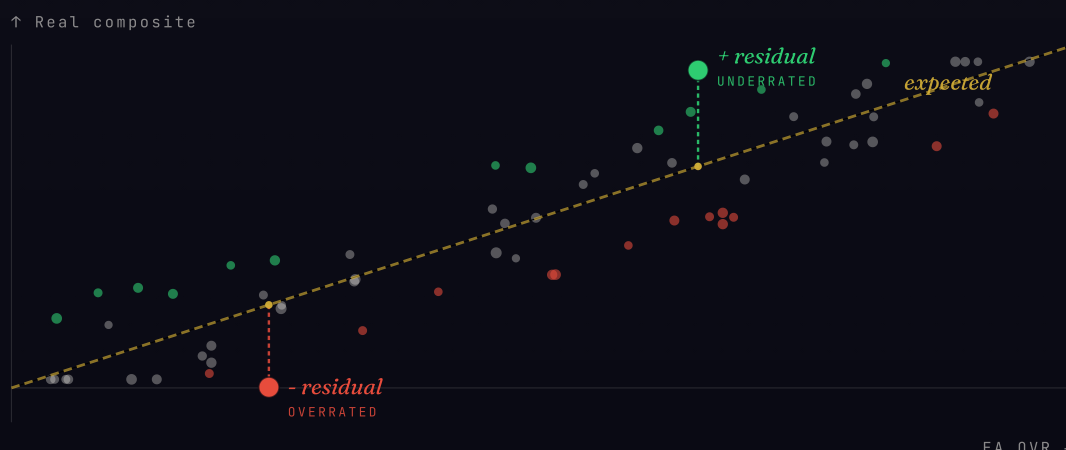
AREA • MINUTES PLAYED

Dot area scales with league minutes, so full-season regulars pull the eye more than rotation players who happen to land at the extremes.

A directional residual, *per sub-position.*

● THE RESIDUAL, VISUALISED

ILLUSTRATIVE SAMPLE • STRIKERS, TOP 5 LEAGUES



● 01 • WHY A RESIDUAL

A delta between OVR and a real metric assumes a one-to-one mapping. There is none, a defender's 0.05 xG/90 is not a striker's 0.05. The signed residual to a fitted line absorbs that distortion and reads on the same axis whatever the role.

● 02 • PER SUB-POSITION

Eight sub-positions, eight regressions. Each line refits when the reader toggles a role or a league, so every dot is judged against the population it actually belongs to. Strikers, central midfielders and centre-backs each get a tailored expectation.

● 03 • THE COMPOSITE

Five dimensions per role. Inside each, FBref metrics are percentile-ranked within the sub-position and averaged with per-role weights. The dimension scores recombine into one composite, then a final z-score rescales it to share OVR's mean and spread.

FIVE DIMENSIONS

SCORING

CREATION

PROGRESSION

DEFENSE

DISCIPLINE

PERCENTILE INSIDE • Z-SCORE OUTSIDE • PER-ROLE WEIGHTS

Two challenges, *three lessons*.

● CHALLENGE 01

Matching player names across sources.

EA writes K. Mbappé, FBref writes Kylian Mbappé Lottin, Transfermarkt drops the accent altogether. We combined rapidfuzz token-sort ratio with a nation and club tiebreaker: above 95 we accept the match outright, between 75 and 95 we keep it only if nation or club agree, the rest falls to a partial-ratio pass within the same nation. Edge cases (Carlos Alcaraz vs Carlos Alvarez) were resolved by hand.

EA	FBref	ratio
K. Mbappé	Kylian Mbappé Lottin	94
Vinicius Jr.	Vinicius Junior	90
A. Griezmann	Antoine Griezmann	100
M. Marín	Marc Marín Madariaga	71 → manual
Lee K.	Kang-In Lee	63 → manual

- 92.9% match rate on the 1,839 regulars ($\geq 600'$). The 131 misses are mostly middle-name variations (Jon Aramburu ↔ Jon Mikel Aramburu); one is genuinely absent from EA.

● CHALLENGE 02

Talking xG to readers who only know OVR.

The audience splits in two. Fans and FC 25 players know what OVR means but have never seen xG; stats readers know npG and SCA90 by heart. We introduce each concept only when it earns its place, and keep definitions one hover away.

PROGRESSIVE DISCLOSURE ACROSS THE THREE ACTS

- ACT I Only OVR and a ranked list of names. The reader sees the rankings disagree before any new metric is introduced.
- ACT II The composite, one regression line per sub-position, residual on the y-axis. A Y-axis switch lets the curious peek at raw xG or goals.
- ACT III Five named dimensions, never raw FBref column names. Sliders speak the same language as the quiz that set them.

- Football phrasing on every label: strikers, wingers, defensive midfielders, never raw FBref codes. Small (i) markers next to every chart, one-paragraph tooltips on hover.

● LESSONS LEARNED

THREE THINGS THAT STAYED WITH US

01 • METHOD

The composite is too linear for goalkeepers.

Sweeping and distribution barely register in FBref's tabular GK stats, so our GK residuals are noisier than the rest. GK should get its own non-linear blend.

02 • CALIBRATION

Diverge from EA, but not into nonsense.

Tuning the composite was a balancing act: enough divergence from OVR to surface a real gap, close enough to football common sense that the result stayed credible. A composite flagging Modrić as overrated would have killed the frame, regardless of how clean the math was.

03 • THE META-LESSON

We set out to prove EA wrong. We learned every rating is.

M1 was built on the assumption that EA was biased and we would expose it. By mid-M2 the deeper point landed: yes EA is biased, our composite is too, and every rating carries the bias of whoever picks the weights. That is the lesson we hope the reader keeps, as much as we did.

ON AI ASSISTANCE

A faster rubber duck, not a co-author. AI was used for narrow tasks: browser debugging and motion polish. The methodology, residual framing, copy and design decisions stayed ours.

Three people, three acts.

One lead per act, the rest shared.

ACT I & DATASET

Youssef Jeddi

SCIPER 355 976

- Act I in full: split-screen ranking, sub-position toggle, animated linking lines.
- Detailed player card: EA sub-ratings, real-stat radars, heatmaps, GK panel, Transfermarkt strip.
- Site intro and dataset presentation: reduction funnel, EA, FBref and Transfermarkt source dossiers.

ACT II & MAP

Omar Bouden

SCIPER 341 381

- Act II in full: Guess the Gap intro, scatter with per-position regression, over/underrated sidebar.
- Reputation Gap league map: continent → country → club drill and the per-club stadium view.
- Hidden gems and reputation tax ledger, and the multi-player radar overlay.

ACT III & POLISH

Alexis Allemand

SCIPER 355 689

- Act III in full: line-up picker, opinion quiz, sliders, presets, live impact panel, taste-test recap.
- Conclusion section, recap of what the reader has just rebuilt and what the project actually measured.
- End-of-project site-wide pass: visual unification across acts, bug fixes, README and process book.

SHARED

Data preparation, overall site design, and this process book were done together by the three of us. Screencast: Omar and Youssef.

● STACK

HTML, CSS, VANILLA JS

D3.JS V7

SCROLLAMA 3

GSAP 3

TOPOJSON CLIENT

PANDAS, NUMPY

RAPIDFUZZ

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